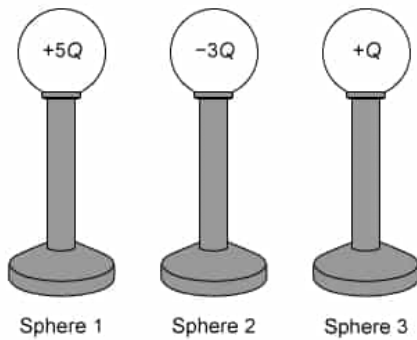


Three identical conducting spheres are mounted on insulating handles. Sphere 1 has a charge of $+5Q$, sphere 2 has a charge of $-3Q$, and sphere 3 has a charge of $+Q$.



There are 24 excess electrons on the surface of sphere 2. What is the electric charge on sphere 1? (The magnitude of electron charge is 1.6×10^{-19} C.)

- ☐ A. 0.96×10^{-18} C [5%]
☐ B. 1.28×10^{-18} C [25%]
☐ C. 4.80×10^{-18} C [30%]
☒ D. 6.40×10^{-18} C [38%]

Correct

39% answered correctly

Explanation:

Using the charge of the electron to determine the total charge on a conducting sphere

- Q Electric charge
- e Fundamental charge
- q_1 Initial charge on sphere 1
- q_2 Initial charge on sphere 2

Sphere 2

The initial charge on sphere 2 is equal to the charge of 24 electrons

$q_2 = -3Q$ $q_2 = 24(-e)$
 $-3Q = -24e$
 $Q = 8e$

Sphere 1

Fundamental charge $e = 1.6 \times 10^{-19}$ C

$Q = 8e$ $q_1 = +5Q$
 $q_1 = +5(8e) = 40e$
 $q_1 = 6.40 \times 10^{-18}$ C

Electric charge is quantized, and an **electron** carries the smallest unit of charge that can be isolated, which is the **elementary charge** $e = \pm 1.6 \times 10^{-19}$ C. Hence, the charge q on an object is equal to the product of the number of electrons n and the fundamental charge:

$$q = \pm ne$$

Thus, the number of electrons on a charged object is:

$$n = \frac{q}{e}$$

In this question, the charge q_2 on sphere 2 is represented both in terms of Q and by the number of excess electrons on its surface ($-3Q$ and $-24e$, respectively). Thus, the value of Q can be determined in terms of e :

$$q_2 = -3Q = -24e$$

$$Q = \frac{-24e}{-3} = 8e$$

The value of Q is equal to $8e$ and can be substituted into the charge of sphere 1, $q_1 = +5Q$, to determine sphere 1's charge as a number of electrons:

$$q_1 = +5Q = 5 \cdot (8e)$$

$$q_1 = 40e$$

Plugging in the elementary charge yields:

$$q_1 = 40 \cdot (1.6 \times 10^{-19}) = 6.40 \times 10^{-18} \text{ C}$$

Therefore, the charge on sphere 1 is equal to $6.40 \times 10^{-18} \text{ C}$.

(Choice A) The charge $0.96 \times 10^{-18} \text{ C}$ results from determining that $Q = 6e$ through incorrect division of $-24e$ and $-3Q$ and without multiplying by 5 to represent the charge on sphere 1.

(Choice B) The charge $1.28 \times 10^{-18} \text{ C}$ results when $Q = 8e$ is not multiplied by 5 to represent the charge on sphere 1.

(Choice C) The charge $4.80 \times 10^{-18} \text{ C}$ results from determining that $Q = 6e$ (through incorrect division of $-24e$ and $-3Q$) and is multiplied by 5 to represent the charge on sphere 1.

Educational objective:

The charge on an electron is the smallest amount of charge that can be isolated in nature. The number of electrons on an object equals the total charge divided by the charge of the electron.

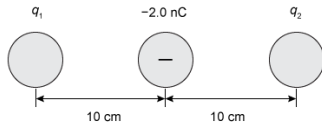
Time spent:0 sec

Subject:
Physics

QID:404141

System:
Electrostatics and Circuits

Two small spheres with charges q_1 and q_2 are 20 cm apart. A third sphere with a charge of -2.0 nC is placed directly between them.



If the sphere with charge q_2 experiences a zero net electric force, what is the value of q_1 ?

- ☐ A. -8 nC [8%]
☐ B. 4 nC [43%]
☒ C. 8 nC [44%]
☐ D. 16 nC [3%]

Correct

45% answered correctly

Explanation:

Calculating the value of an unknown charge using Coulomb's law

$F_E = k \frac{|q_A| \cdot |q_B|}{r^2}$

- F_E Electric force
- k Coulomb constant
- q_A, q_B Charge magnitudes
- r Separation distance
- $F_{E,1}$ Electric force due to q_1
- $F_{E,2}$ Electric force due to -2.0 nC charge

Zero net force on q_2

$$F_{E,1} = F_{E,2}$$

$$k \frac{|q_2| \cdot |q_1|}{(20 \text{ cm})^2} = k \frac{|q_2| \cdot (2.0 \text{ nC})}{(10 \text{ cm})^2}$$

$$q_1 = 8.0 \text{ nC}$$

Coulomb's law implies that the magnitude of the electrical force F_E between two charged objects (A and B) is **directly proportional** to the product of the charge magnitudes $|q_A|$ and $|q_B|$, and **inversely proportional** to their squared separation distance r^2 :

$$F_E = k \frac{|q_A| \cdot |q_B|}{r^2}$$

The electric constant k is equal to $8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2$. The direction of the electric force depends on the **sign of the charges**: like charges repel, and opposite charges attract.

In this question, the sphere with charge q_2 experiences **zero net electric force**. For the electric forces to sum to zero, q_1 must be positive. It follows that the electric force exerted on q_2 due to q_1 ($F_{E,1}$) is always directed opposite to the force exerted on q_2 due to the -2.0 nC charge ($F_{E,2}$). Furthermore, the magnitude of the forces must be equal:

$$F_{E,1} = F_{E,2}$$

$$\frac{k \cdot |q_2| \cdot |q_1|}{(20 \text{ cm})^2} = \frac{k \cdot |q_2| \cdot (2.0 \text{ nC})}{(10 \text{ cm})^2}$$

The values of k and q_2 cancel out on both sides of the equation. Solving for q_1 yields:

$$q_1 = \left(\frac{20 \text{ cm}}{10 \text{ cm}} \right)^2 \cdot 2.0 \text{ nC}$$

$$q_1 = 4 \cdot 2.0 \text{ nC} = 8.0 \text{ nC}$$

Therefore, the value of q_1 is 8.0 nC.

(Choice A) -8.0 nC has an incorrect sign. If the value of q_1 is negative, then $F_{E,1}$ and $F_{E,2}$ would not be oppositely directed and q_2 would experience a nonzero net force.

(Choice B) 4.0 nC results from not using the square of the separation distances in the calculations with Coulomb's law.

(Choice D) 16.0 nC follows if an extra factor of 2 is incorrectly included by considering the middle -2.0 nC charge twice.

Educational objective:

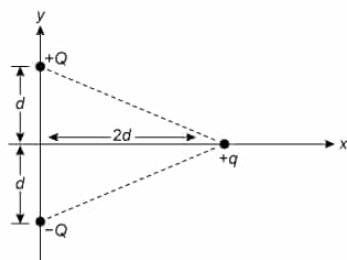
Coulomb's law states that the electric force between two charges is directly proportional to the magnitude of each charge and inversely proportional to the square of the separation distance between the charges. A zero net force condition requires equal and oppositely directed forces.

Time spent:0 sec

Subject:
Physics

QID:404142

System:
Electrostatics and Circuits



Three charged objects are in an xy -plane, as shown above. Charges $+Q$ and $-Q$ are located at positions $(0, d)$ and $(0, -d)$, respectively. Charge $+q$ is located at position $(2d, 0)$. All three charges are held fixed in place. What is the magnitude of the electric force that is exerted on charge $+q$ by each charge $+Q$ and $-Q$?

- ☐ A. $\frac{kqQ}{d^2}$ [13%]
- ☐ B. $\frac{kqQ}{4d^2}$ [37%]
- ☒ C. $\frac{kqQ}{5d^2}$ [42%]
- ☐ D. $\frac{kqQ}{9d^2}$ [6%]

Correct

43% answered correctly

Explanation:

Using Coulomb's law to determine the electrostatic force between charged objects

- q, Q Electric charge
- k Electric constant
- r Separation distance
- d Distance
- F_E Electric force

Coulomb's Law

$$F_E = k \frac{|q_1||q_2|}{r^2}$$

Charges $+Q$ and $-Q$ each exert the same electrostatic force on $+q$ because they have the same charge and are separated by the same distance.

Pythagorean Theorem

$$r^2 = d^2 + (2d)^2$$

$$r = \sqrt{d^2 + (2d)^2} = \sqrt{5d^2}$$

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Coulomb's law implies that the magnitude of the electrical force F_E between two charged objects (A and B) is **directly proportional** to the product of the charge magnitudes $|q_A|$ and $|q_B|$, and **inversely proportional** to their squared separation distance r^2 :

$$F_E = k \frac{|q_A| \cdot |q_B|}{r^2}$$

The electric constant k is equal to $8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2$. The direction of the electric force depends on the **sign of the charges**: like charges repel, and opposite charges attract.

In this question, charges $+Q$ and $-Q$ have the same magnitude of charge and are located at equal distances away from charge $+q$. Hence, the electric force exerted on $+q$ by $+Q$ is equal in magnitude to the electric force exerted on $+q$ by $-Q$. The separation distance r can be found from the **Pythagorean theorem**:

$$r^2 = d^2 + (2d)^2$$

$$r = \sqrt{d^2 + (2d)^2} = \sqrt{5d^2}$$

From Coulomb's law, the electric force that charge $+Q$ and $-Q$ each exert on charge $+q$ is equal to:

$$F_E = k \frac{|q_A| \cdot |q_B|}{r^2} = \frac{kqQ}{(\sqrt{5d^2})^2}$$

Therefore, the correct answer is Choice C:

$$F_E = \frac{kqQ}{5d^2}$$

(Choices A, B, and D) The forces $\frac{kqQ}{d^2}$, $\frac{kqQ}{4d^2}$, and $\frac{kqQ}{9d^2}$ result from incorrectly assigning the distance separation between the charges q and $\pm Q$ as d , $2d$, or $3d$, respectively.

Educational objective:

Coulomb's law states that the electric force between two charges is directly proportional to the magnitude of each charge and inversely proportional to the square of the separation distance between the charges. Distance can be determined using the Pythagorean theorem.

Time spent:0 sec
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Physics

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Electrostatics and Circuits